Scenario: In a frontend interview, Jane was tasked with creating a timer user interface that displays the elapsed time in seconds, incrementing every second. The interface should include three buttons: Start, Pause, and Reset. The timer should begin counting up when Start is clicked, pause when Pause is clicked, and reset to zero when Reset is clicked. After implementing the timer, Jane was also asked to explain the key JavaScript concepts used in her solution.

Analysis: This scenario serves as a practical assessment of both fundamental and advanced JavaScript skills, especially within the context of modern frontend frameworks. It evaluates a candidate's ability to translate requirements into a functional, interactive component while demonstrating proficiency with contemporary JavaScript and React concepts. Additionally, it tests not only technical expertise but also the ability to communicate clearly and clarify requirements—skills that are essential for effective collaboration and success in real-world development.

Jane's Approach: When she encountered this question, she clarified with the interviewer that she would design a UI similar to figure 3.3, showing the elapsed time (which increases every second) and three buttons: Start, Pause, and Reset. The timer should begin counting up when Start is clicked, pause when Pause is clicked, and reset to zero when Reset is clicked. The interface should visually reflect the timer's state—for example, after starting, the timer might show an elapsed time such as 21 seconds.

Figure 3.3 Timer user interface after starting a 21-second countdown